Acuity Engine V3 Review

Bottom line: the assignment problem grows combinatorially, so humans cannot reliably evaluate more than a tiny fraction of the available legal boards. A constrained optimizer can score hundreds to thousands of candidate states consistently inside a short time budget and can re-run after every staffing or patient change.

1. Problem Scale

If `P` patients can each be assigned to one of `N` owners, the raw assignment space is approximately `N^P` before constraints, locks, and roster exclusions remove impossible combinations.

Example	Approximate search space	Interpretation
8 RNs, 32 patients	`8^32 ≈ 7.9 × 10^28`	Far beyond any human mental search process and far beyond direct brute force.
4 PCAs, 32 patients	`4^32 ≈ 1.84 × 10^19`	Still enormous. Requires pruning, scoring, and bounded search.

2. Why Software Has an Advantage

What a human does well

- Spots obviously unsafe or unfair assignments quickly.
- Uses contextual judgment the system may not yet model.
- Balances clinical nuance when the rules are ambiguous.

What the optimizer does better

- Checks the same rules the same way every time.
- Does not forget a constraint after an interruption.
- Can compare many alternative boards inside one rebalance cycle.
- Can re-run after every staffing, acuity, or discharge-status change.

Practical interpretation: a charge nurse may mentally compare a handful of candidate layouts while also taking interruptions. The engine can score many more candidate boards in a bounded search window while consistently preserving explicit constraints.

3. Version Progression

Version	Main style	Primary strength	Main limitation
V1	Heuristic populate + repair	Gets to a workable board quickly	Becomes patch-heavy as more priorities are added
V2	Scored optimizer with local improvements	Compares boards more coherently than V1	Can get stuck in decent but not best local optima
V3	Multi-seed + beam-search optimization	Searches multiple promising board families before committing	Still bounded, so not a full exhaustive global optimum

4. Current V3 Priority Stack

1. Safe assignment with no preventable rule breaks
2. Balanced patient counts
3. Balanced acuity / load spread
4. Minimal report-source burden
5. Tighter room clustering

5. Expected Efficiency Impact

These are operational expectations, not outcome guarantees.

- Faster regeneration of candidate boards after data changes.
- More consistent enforcement of explicit staffing rules.
- Reduced chance of preventable oversights caused by interruption or working-memory overload.
- More repeatable fairness decisions across shifts and users.

6. Downstream Safety Value

The platform does not remove clinical risk. It reduces one specific source of operational risk: preventable assignment inconsistency.

- Fewer preventable stacking patterns should reduce exposure to unfair workload concentration.
- More consistent balancing should reduce the chance that one owner receives a hidden high-burden cluster.
- Lower dependence on manual mental recomputation should reduce error opportunities during interruptions.

Important note: the statements above are reasoned operational inferences from better constraint consistency. They are not claims of proven patient-outcome effect unless validated by real unit data.

7. Why This Matters

Acuity assignment is not just a formatting problem. It is a constrained decision problem with a search space too large for manual comparison. The engine’s value is that it can repeatedly search, score, and improve assignments under pressure while keeping the same explicit priorities every time.

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